

Research Paper Overview

ECON 4341W Poverty and Income Inequality

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Structure of an Empirical Paper

- ① Introduction
- ② Literature Review
- ③ Data
- ④ Methodology/Empirical Model
- ⑤ Results
- ⑥ Conclusion
- ⑦ Appendix

Topic Selection

Start selecting your topic from:

- ① class material
 - Lecture and reading could be very helpful!
 - The contents of assigned textbooks
 - Any discussions or papers mentioned that interest you?
- ② current policy debate/real-world issue that you care about

From the Topic to a Specific Question

Try to start by framing your research question like this:

$$Y = X\beta + \epsilon.$$

- What is on the left-hand side?
- What is the key right-hand-side variable of interest?
- What are some other possible confounding variables on the right-hand side?

» Example

» Regression

Data: Where to Find them?

① Aggregate Data

- Canvas Module
 - Gini Index, Poverty Threshold, Median Income by state/by race, ...
- “data and methodology” section in your reading
- FRED (Federal Reserve Economic Data)
- Our World in Data

② Microdata

- IPUMS:
 - ACS (American Community Survey)
 - CPS (Current Population Survey)
- World Bank Microdata Library
- The DHS (Demographic and Health Surveys) Program
- ICPSR (Inter-University Consortium for Political and Social Research)

Data: How to understand them?

Things to check:

- The unit of observation, the number of observations.
- Summary statistics of variables, especially the variables relevant to your questions (LHS, RHS, and confounding variables)

Tips:

- 1 Always carefully check the codebook and documentation that come with the data.
- 2 Interact with the data with the tools that you know.

► Details

Regression: Correlation to Causality

- “Methodology/Empirical Part” in an Empirical Paper
- Endogeneity Problem
 - The right-hand-side variable is correlated with the error term, leading to biased estimates.
 - Hence the way we frame our question... [» Question](#)
- Objective: Deal with the endogeneity problem [» Resources](#)
 - Selection on observables
 - Difference-in-differences
 - Instrumental Variable
 - Regression Discontinuity

An Example

Topic: Family Economics

- What are the economic roles of the households?
- How does the role shape individual's marriage decision?

Question: What is the effect of the Great Recession (2007-2009) on marital status and Marital sorting? [» Format](#)

- LHS: Marital Status and Marital Sorting
- Key RHS variable: the state-level unemployment percentage rate change during the recession period
- Possible confounding variables on RHS
 - age, education, ...

Topic → Question

① Theoretical-driven

- Economic roles of the household: risk-sharing behavior?
- The Great Recession as a potential exogenous shifter of labor income risk that individuals are facing.
- What describes individual's marriage decision?
- Extensive margin: marriage/divorce
- Intensive margin: sorting

② Literature

- Rich literature on the effect of the Great Recession. It might be interesting to explore ...

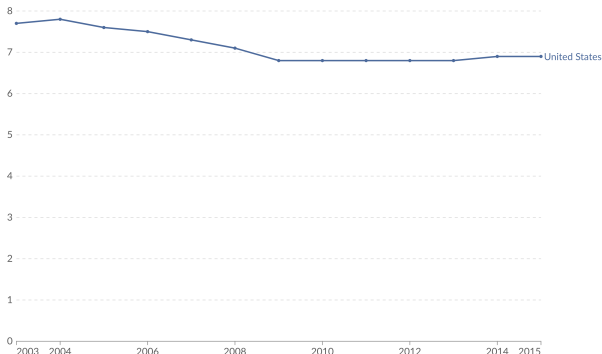
③ Interacting with aggregate data

Interacting with Aggregate Data

Marriage rates

Number of marriages during a given year per 1,000 people.

Our World
in Data



Data source: OECD (2025) and other sources

OurWorldinData.org/marriages-and-divorces | CC BY

Figure: Marriage Rates, U.S, 2003-2015

Given the question, interact with data?

- ACS (American Community Survey), 2003-2015
 - If you are using survey data, please be careful about the survey weight.
- See details in the HTML file.

Logistics

- Let's meet!
- Writing Assistant: Yiyuan He
- Office Hour: Friday 2-5 PM (by appointment)
- Office Hour Registration Link:
<https://calendar.app.google/vxJM6Bmwj5mucGPB8>
- Office Hour in person: Hanson 3-107
- Office Hour Zoom Link: <https://umn.zoom.us/j/2124031679>

Data Type

- **Microdata** are the “raw” data, composed of individual records containing information collected on the research subjects being studied. The unit of observation is usually the individual, but can be the household, family, etc.
- **Aggregate data** are compiled statistics, such as a table of marital status by sex that summarizes the responses obtained from a survey.

► Data Source

Data: How to understand them?(Detailed)

- 1 What is the unit of observation in your data?
- 2 How many observations are in your dataset?
- 3 How many variables are in your dataset?
- 4 What is the single most important variable that will be on the left-hand-side of your regression?
- 5 What is the key right-hand-side variable (or variables) most important for investigating your hypothesis?
- 6 What other variables do you plan to put on the right-hand-side in order to control for confounding influences?
- 7 What is the mean of your left-hand-side variable? What is the range of this variable?
- 8 Give the same information for the right-hand-side variables.
- 9 What is the correlation between your left-hand-side variable and the key right-hand-side variable?

Regression: Helpful Resources

Mostly Harmless Econometrics by Angrist and Pischke

- Chapter 2: A more modern framework/approach to understand endogeneity problem
- Chapter 3: Introduction to linear regression and selection on observables
 - For the purpose of doing your own project, you can skip the proof and the discussion about matching.
- Other chapters: depend on which model fits your question.

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